

Vijay Kumar

Software Engineer | Java Backend & Real-Time Systems

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PROFESSIONAL SUMMARY

Backend engineer with production experience building real-time distributed systems using **Java 17** and **Spring Boot 3**. Designed and shipped **Nexus** — a live multiplayer game platform with STOMP WebSocket sessions, PostgreSQL persistence, bcrypt authentication, and automated CI/CD via GitHub Actions to Oracle Cloud. Strong foundation in concurrency, event-driven design, and systems that hold up under real user load. Targeting remote backend engineering roles.

TECHNICAL SKILLS

Languages	Java 17, JavaScript (ES2022), SQL, HTML / CSS
Backend	Spring Boot 3, Spring Data JPA, WebSocket (STOMP/SockJS), RESTful APIs, Maven
Databases	PostgreSQL (Neon), H2 (testing), JPQL bulk queries, Hibernate ORM 6
Infrastructure	Docker, GitHub Actions (CI/CD), Linux (Ubuntu 22.04), Vercel, Render, Oracle Cloud VM
Architecture	LLD / HLD, OOP, Multithreading & Concurrency, Event-Driven Design
Fundamentals	Data Structures & Algorithms, Operating Systems, Computer Networks

TECHNICAL PROJECTS

Nexus — Real-Time Multiplayer Game Platform

Java 17 · Spring Boot 3 · WebSocket · PostgreSQL

[nexusgame.space](#) | [github.com/VijayKumarCode/nexus](#)

- › Designed and deployed a full-stack real-time multiplayer game platform; currently live with concurrent user sessions managed via STOMP WebSocket and SockJS.
- › Built a stateful game engine using in-memory ConcurrentHashMaps for toss, challenge, and move state; added DB-backed recovery so active games survive server restarts.
- › Diagnosed Railway port-587 SMTP blocking through production log analysis; replaced JavaMailSender with Resend HTTP API (port 443) — activation email latency from 2-minute timeout to under 1 second.
- › Fixed JPQL enum syntax failures (User\$UserStatus inner-class reference) causing scheduled tasks to crash every 10 s, silently rejecting all game moves.
- › Secured API with @JsonIgnore on password / activation-token fields; closed XSS surface in lobby via data-attribute event delegation replacing inline onclick interpolation.
- › Configured GitHub Actions CI/CD: builds JAR, copies via SCP to Oracle Cloud VM, restarts systemd service — zero-touch deployment on every commit to main.

TicTacToe — Network Multiplayer Engine

Java · Sockets · Multithreading

[github.com/VijayKumarCode/TicTacToe_java](#)

- › Built a socket-based two-player game server managing concurrent client sessions with dedicated threads per connection and synchronized shared game state — architectural predecessor to the Nexus platform.
- › Implemented server-side win detection, move validation, and real-time board synchronization across two connected clients.

DSA In Java — Algorithm & Data Structure Library

Java 17 · Algorithms

[github.com/VijayKumarCode/Java-Data-Structures-and-Algorithms](#)

- › Implemented and documented Trees, Graphs, Linked Lists, Hash Maps, and sorting algorithms in Java 17 with verified time and space complexity for each solution.

Portfolio Website

HTML · CSS · JavaScript · Vercel

[vijaykumarcodespace](#) | [github.com/VijayKumarCode/Vijay-Kumar-Portfolio](#)

- › Built with vanilla HTML, CSS, and JavaScript — no frameworks. Deployed on Vercel with a custom domain and continuous deployment on every push to main.
- › Spring Boot 3 backend on Render handles contact form persistence (PostgreSQL/Neon) and email notification via Resend HTTP API.

EDUCATION

Master of Computer Applications (MCA)

AKGEC, Ghaziabad, Uttar Pradesh

Bachelor of Computer Applications (BCA)

LNMCBM, Muzaffarpur, Bihar